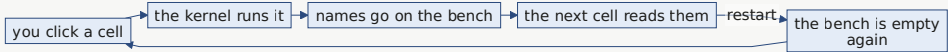


1 the workbench



The kernel is a workbench that holds whatever you put on it for as long as it is running.

The move	What it does to the bench	What it does to the file
You run a cell.	It puts a name and a value on the bench.	It changes nothing on disk.
You restart the kernel.	It sweeps the bench completely clean.	It changes nothing on disk.
You restart and run all.	It rebuilds the bench from the top in a few seconds.	It changes nothing on disk.

The notebook file and the kernel state live in two different places, and only one of them is erased by a restart.

Crux. a restart costs you the state and never the file, which is why the recovery is cheap.

2 cells run in the order you ran them

The kernel sees your run history and has no opinion at all about where a cell sits on screen.

```
[2] records = [ ... ]
[3] count = 0
[1] for r in records:
```

The cell at the bottom ran first, and the counter in the square brackets is the only honest record of what happened.

```
NameError: name 'records' is not defined
```

You asked for a name the bench was not holding, and the message names the exact word it went looking for.

Crux. when an output surprises you, read the execution counters before you read the code.

3 four types, and the question you ask

Type	What it holds	From today's file
str	It holds text, including text made only of digits.	The id "KR4224" and the amount "4500" on KR4200.
int	It holds a whole number you can do arithmetic with.	The amount 1460 on KR4224.
float	It holds a number with a decimal part.	A value such as 1460.5.
bool	It holds True or False, which every comparison hands back.	The result of 2395 > 2000.

```
type("4500")      <class 'str'>
type(4500)        <class 'int'>
type(4500 > 2000) <class 'bool'>
```

Crux. ask a value what it is before you change any code, because the quotes are the easiest character on the line to skip.

4 the comparison Python refuses

```
if r["amount"] > 2000:

TypeError: '>' not supported between instances of 'str' and 'int'
```

The loop stops on KR4200, the first card in the file, whose amount is the text "4500".

The fix converts at the point of use, so the record stays as it is and the comparison gets a number.

```
if int(r["amount"]) > 2000:
```

With that one change the cell walks all thirty records and reports 13 orders above Rs 2,000 totalling Rs 35,020.

Crux. Python refuses rather than guessing, and a spreadsheet would have handed you a number with no warning attached to it.

5 the loop, the condition and the accumulator

```
count = 0
total = 0
for r in records:
    if r["status"] == "delivered":
        count = count + 1
        total = total + int(r["amount"])

print(count, total)
```

13 25720

The zeros sit above the loop and inside the same cell, so rerunning the cell starts from zero instead of adding a second copy onto the first run.

Move `total = 0` inside the loop and the cell still runs, printing 1460, which is the amount of KR4224 and a total of nothing at all.

Crux. a total over a group of orders can never be smaller than the largest order in that group, so check the number before you trust the code.

6 lists, and the second name

```
ids = ["KR4200", "KR4201", "KR4202"]

ids[0]      "KR4200"
ids[1:3]    ["KR4201", "KR4202"]
```

Counting starts at zero, and a slice stops before its second number.

```
a = ["KR4200", "KR4201"]
b = a
b.append("KR4202")

len(a)      3
```

`b = a` gives one list a second name, so appending through either name changes what both names show. Write `b = list(a)` when you want a second list on purpose.

Crux. one list with two names is one list, and the bug it causes shows up far from the line that caused it.

7 records, the missing key and the default

```
r["status"]      "delivered"
r["amount"]      1460
records[0]["discount"]  KeyError: 'discount'
```

Two of the thirty orders carry a `discount` field and twenty-eight carry no such field at all, so absent is a different thing from empty.

```
r.get("discount", 0)
```

The default you write	What the thirty records total	What it claims
<code>r.get("discount", 0)</code>	The discounts total Rs 250.	It claims that a record with no discount field was not discounted.
<code>r.get("discount", 100)</code>	The discounts total Rs 3,050.	It claims a discount of Rs 100 on twenty-eight orders that never recorded one.

Both cells run to the end and both print a clean number.

Crux. the default is a decision you own, and it leaves no mark on the screen once the cell has run, so you have to say it out loud.

8 reading an error in three questions

The question	Where you find the answer
What is the exception type?	It is the first word on the last line of the message.

Which line is mine? It is the line the message points at in your own cell.

What value was it holding? It is the thing named in quotes or described by type at the end of the message.

The third question is the one people skip, and it is the one that names the record.

Today's error	What it was holding
<code>NameError: name 'records' is not defined</code>	It was holding nothing, because the setup cell had not run.
<code>TypeError: '>' not supported between instances of 'str' and 'int'</code>	It was holding the text "4500" from KR4200 on one side.
<code>KeyError: 'discount'</code>	It was holding a record that has no field of that name.

Crux. every one of these names the thing it choked on, so read it before you edit anything.

THE WORDS ON THIS SHEET

Kernel The Python that runs your cells and holds every name you gave it, until it...

Execution counter The number in square brackets, counting runs rather than positions on the...

Type What a value is, which decides what every operator means for it

Accumulator A name set once above a loop and changed on every qualifying turn

NameError The kernel was asked for a name nothing had put on the bench

TypeError An operator was handed two values it has no honest meaning for

KeyError A dictionary was asked for a name that record does not carry

Default The value you supply when a field is absent, which is a claim about the...